Half-integral continued fractions for Chebyshev products

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Abstract

We prove Cohn's Conjecture 9.2: for every nonzero multiple ℓ of four, the product $\prod_{n\geq 0}(1+1/T_{\ell^n}(x))$ has a semi-specializable continued fraction. We give an explicit recursion that preserves integral coefficients at odd positions and half-integral coefficients at even positions. The argument applies more generally to every nonconstant $f \in \mathbb{Z}[x]$ satisfying $f(-1) = f(0) = f(1) = 1$.

Task 1: statement and a general theorem

Work in the field $\mathbb{Q}((x^{-1}))$, with convergence measured by degree at infinity. A regular continued fraction has polynomial partial quotients, all of positive degree after its initial term. Following Cohn [1, Sections 7 and 9], semi-specializability means that all their coefficients have denominators at most two, while the fraction is not specializable over $\mathbb{Z}[x]$. The target is Conjecture 9.2 of that paper, for the Chebyshev polynomials T_m.

Theorem 1. *Let $f \in \mathbb{Z}[x]$ be nonconstant and satisfy*

$$f(x) = 1 + x(x-1)(x+1)g(x), \qquad g \in \mathbb{Z}[x]. \tag{1}$$

Set $f_0 = x$, $f_{n+1} = f(f_n)$, and

$$R_n = \prod_{j=0}^{n} \left(1 + \frac{1}{f_j}\right).$$

The regular continued fractions of the R_n form nested prefixes of an infinite continued fraction $[1; a_1, a_2, \ldots]$ for R_∞. For every $j \geq 1$,

$$a_j \in \mathbb{Z}[x] \quad (j \text{ odd}), \qquad 2a_j \in \mathbb{Z}[x] \quad (j \text{ even}).$$

Its first four partial quotients after 1 are

$$x, \quad -(x-1)g(x), \quad 1-x, \quad -1-\frac{x}{2}. \tag{2}$$

In particular it is semi-specializable.

The hypotheses force $g \neq 0$ and $d := \deg f \geq 3$. For $\ell \neq 0$ divisible by four, $T_{|\ell|}(0) = T_{|\ell|}(1) = T_{|\ell|}(-1) = 1$. The monic factor theorem over $\mathbb{Z}[x]$ gives (1). The identities $T_a(T_b(x)) = T_{ab}(x)$ and $T_{-a} = T_a$ identify f_n with T_{ℓ^n}. Thus Theorem 1 proves the full conjecture, including negative ℓ.

A continued-fraction identity

For a polynomial a write

$$A(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

For a word $w = (a_1, \ldots, a_m)$ with $a_1 = x$, put

$$M = A(1)A(a_1)\cdots A(a_m) = \begin{pmatrix} p & p' \\ q & q' \end{pmatrix}, \qquad \epsilon = \det M = (-1)^{m+1}.$$

Here p, q are the actual continuants, including their scalar factors. Define the word $\mathcal{T}(w, Y)$ as follows:

$$\mathcal{T}(w, Y) = \big(w, Y, -a_m, \ldots, -a_2, 1 - x, b_1, \ldots, b_m\big), \tag{3}$$

where

$$b_1 = -1 - \frac{x}{2}, \qquad b_j = \begin{cases} -2a_j, & j \geq 2 \text{ even}, \\ -a_j/2, & j \geq 3 \text{ odd}. \end{cases}$$

When $m = 1$, the segment $-a_m, \ldots, -a_2$ is empty. The new word has length $3m + 1$ and begins with w.

Lemma 1. *Let F satisfy $Y = -\epsilon(F-1)/(pq)$ and set*

$$c_m = \begin{cases} 1, & m \text{ even}, \\ 1/2, & m \text{ odd}. \end{cases}$$

The first column of the continuant matrix of $[1; \mathcal{T}(w, Y)]$ is

$$-\epsilon c_m \begin{pmatrix} p(F+1) \\ qF \end{pmatrix}. \tag{4}$$

Consequently, this fraction has value $\dfrac{p}{q}(1 + 1/F)$.

Proof. Let $W = A(a_1)\cdots A(a_m)$ and $D = \operatorname{diag}(1, -1)$. Then

$$W = \begin{pmatrix} q & q' \\ p - q & p' - q' \end{pmatrix}.$$

Write S for the continuant matrix of the scaled word $(a_1/2, 2a_2, a_3/2, 2a_4, \ldots)$, and let $e_1 = (1, 0)^{\mathsf{T}}$. Pairwise multiplication of the matrices $A(a_j)$ gives

$$Se_1 = c_m \begin{pmatrix} q \\ 2(p - q) \end{pmatrix}.$$

The negative reversed word has matrix $(-1)^m DW^{\mathsf{T}}D$. Consider the auxiliary word

$$[1; w, Y, -\operatorname{rev}(w), 1, a_1/2, 2a_2, a_3/2, \ldots].$$

Eliminating the displayed constant 1 by

$$[a, 1, b, c, \ldots] = [a + 1, -b - 1, -c, \ldots]$$

produces (3). At the matrix level this multiplies the first column by $(-1)^m$, cancelling the sign from the negative reversal. Hence the new first column is

$$MA(Y)DW^{\mathsf{T}}DA(1)Se_1.$$

Using $pq' - p'q = \epsilon$, its last factors give

$$DW^{\mathsf{T}}DA(1)Se_1 = c_m \begin{pmatrix} pq \\ -pq' - \epsilon \end{pmatrix}.$$

Multiplication by $MA(Y)$ therefore yields

$$c_m \begin{pmatrix} p(pqY - 2\epsilon) \\ q(pqY - \epsilon) \end{pmatrix} = -\epsilon c_m \begin{pmatrix} p(F+1) \\ qF \end{pmatrix},$$

as asserted. The constant-elimination sign also follows directly from

$$A(a+1)A(-b-1) = A(a)A(1)A(b)\,\mathrm{diag}(-1,1), \qquad A(-c) = -DA(c)D.$$

□

Induction and coefficient denominators

Put

$$U_n = \prod_{j=0}^{n}(f_j + 1), \quad V_n = \prod_{j=0}^{n} f_j, \quad s_n = \sum_{j=0}^{n} d^j.$$

Iterating (1) gives the exact polynomial identity

$$f_{n+1} - 1 = (x-1)U_nV_n \prod_{j=0}^{n} g(f_j). \tag{5}$$

Indeed, $f_{j+1} - 1 = (f_j - 1)f_j(f_j + 1)g(f_j)$, which telescopes.

Starting with $w_0 = (x)$ and $\kappa_0 = 1$, we inductively construct words w_n of length m_n whose continuants are

$$p_n = \kappa_n U_n, \qquad q_n = \kappa_n V_n, \qquad \kappa_n \in \{\pm 2^{-h} : h \geq 0\}. \tag{6}$$

The base fraction is $R_0 = [1; x]$. Given w_n, define

$$Y_n = (-1)^{m_n} \frac{f_{n+1} - 1}{p_n q_n}, \qquad w_{n+1} = \mathcal{T}(w_n, Y_n). \tag{7}$$

By (5) and (6), $Y_n \in \mathbb{Z}[x]$: the scalar κ_n^{-2} is an integer. Moreover,

$$\deg Y_n = d^{n+1} - 2s_n = \frac{(d-3)d^{n+1} + 2}{d-1} > 0. \tag{8}$$

Lemma 1 proves that $[1; w_{n+1}] = R_{n+1}$ and that

$$\kappa_{n+1} = (-1)^{m_n} c_{m_n} \kappa_n.$$

This preserves the asserted form of κ_n. More explicitly,

$$m_n = \frac{3^{n+1}-1}{2}, \qquad \kappa_n = (-1/2)^{\lceil n/2 \rceil}.$$

It remains to check the coefficient assertion and regularity. Assume that odd positions of w_n have integral coefficients and even positions have half-integral coefficients. In the reversed segment of (3), the entry coming from position j occupies position $2m_n + 2 - j$, which has the same parity as j. The new entry Y_n is integral. The entry $1-x$ occupies the odd position $2m_n+1$. The entry b_j occupies position $2m_n + 1 + j$, of opposite parity to j. Thus the factors 2 and $1/2$ in the definition of b_j give exactly the required coefficient bounds at these new positions. The special $b_1 = -1 - x/2$ also has half-integral coefficients. Every inherited entry has positive degree, as do $1 - x$, b_1, and Y_n by (8). This proves all the induction assertions.

Since all partial quotients after 1 have positive degree, these are regular polynomial continued fractions. Their nested words have lengths tending to infinity. Such a continued fraction converges in $\mathbb{Q}((x^{-1}))$: the degrees of its convergent denominators strictly increase and the difference between consecutive convergents is $\pm 1/(q_j q_{j-1})$. Its subsequence $[1; w_n] = R_n$ tends to the formal product R_∞, since $\deg f_n = d^n \to \infty$. Their limits are therefore equal, proving the coefficient assertion for the infinite product.

The first recursion gives (2). The first nonintegral coefficient occurs in the nonconstant term of $-1 - x/2$. This also rules out specializability by the criterion in [1, Section 7, p. 308]: when the first nonintegral partial quotient has a nonintegral nonconstant coefficient, inserting constant partial quotients cannot make the expansion integral. Thus the conclusion has precisely Cohn's meaning of semi-specializability. This completes the proof of Theorem 1.

Verification and proposed follow-up tasks

The accompanying exact verifier obtains continued fractions by direct Euclidean division of U_n and V_n, and compares them to the recursion. It checks the actual continuant scale and every coefficient's parity condition. The checks cover 48 finite products from ten polynomials, including T_4, T_8, T_{12}, T_{16}, with maximum denominator degree 4,369. They supplement the proof for all n and all eligible f.

Task 2 (proposed). Classify cubic $f \in \mathbb{Z}[x]$ for which $\prod_{n\geq 0}(1 + 1/f_n(x))$ is semi-specializable, including the classes outside (1).

Task 3 (proposed). Turn (3) into explicit recursions for the positive integer partial quotients after specializing $x = M \geq 2$, for each fixed $\ell \equiv 0 \pmod 4$.

Task 4 (proposed). Determine the irrationality exponent of $\prod_{n\geq 0}(1 + 1/T_{\ell^n}(M))$ for every integer $M \geq 2$ and positive $\ell \equiv 0 \pmod 4$.

Task 5 (proposed). For integer $c \neq 0$, find congruence conditions on $f \in \mathbb{Z}[x]$ that give a uniform bound on the coefficient denominators in the continued fraction of $\prod_{n\geq 0}(1 + c/f_n(x))$.

References

[1] H. Cohn, *Symmetry and specializability in continued fractions*, Acta Arithmetica **75** (1996), 297–320, especially Conjecture 9.2, p. 319. `https://doi.org/10.4064/aa-75-4-297-320`. Author manuscript: `https://arxiv.org/abs/math/0008221`.

[2] J. Mc Laughlin, *Symmetry and specializability in the continued fraction expansions of some infinite products*, Journal of Number Theory **127** (2007), 184–219. https://doi.org/10.1016/j.jnt.2007.04.007.

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